

Multiscale mechanics of soft tissues

Soft tissues (skin, cornea, tendon...) play an important role in the architecture of the human body. In particular, they provide a support function linked to their mechanical properties. These can change depending on the environment, but also with age or pathologies.

Soft tissues are essentially made up of a network of collagen fibers (and sometimes elastin), surrounded by a matrix of proteins rich in water. In the end, there are only a few cells. Still, it is these cells which induce the mechanical modifications of the tissue, by reorganizing the network of fibers which surrounds them in reaction to the mechanical movements of this same network. Thus, it seems important to be able to predict how macroscopic mechanical loadings are transmitted at the microscopic scale, and vice versa.

Using a new experimental approach to observe microstructure during mechanical loadings, we were able to show that classical models do not correctly predict the fiber network movements, which led us to construct a new interpretation of the physical origin of fiber motions.

New Constitutive Modeling of Plastic Anisotropy: from Single-crystal to Polycrystalline Response

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Abstract: In this paper, we present a new polycrystalline modeling approach which makes use for the description of the plastic behavior of the constituent grains of the new yield criterion for single-crystals developed by Cazacu et al. [2017]. Using this new single-crystal yield criterion, we predict the influence of individual texture components on the overall anisotropy in plastic properties of polycrystalline metallic sheets. Examples are given for various engineering materials. The high fidelity of the predictions of the anisotropy in both yield stresses and strain-rate ratios demonstrates the strength of the approach. Another added advantage is that once the single crystal behavior is known, no additional calibration or macroscopic mechanical tests are needed in order to predict the effect of different texture components on the polycrystalline behavior.

Reference:

Cazacu, O., Revil-Baudard, B., Chandola, N. 2017, “A yield criterion for cubic single crystals”, Int J. Solids Struct., DOI: 10.1016/j.ijsolstr.2017.04.006.

Dislocation Dynamics Simulations of Materials with Complex Physics

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Discrete dislocation dynamics (DDD) simulations provide a technique for examining the effects of fundamental dislocation physics on the plastic response of crystalline solids. Many DDD simulations focus on relatively simple materials and loading conditions, such as glide-motion-dominated plasticity of pure face-centered and body-centered cubic crystals. In this presentation, we provide an overview of the more complex physical aspects of dislocation-mediated plasticity in the context of DDD. We consider both physics that are intrinsic to the crystal lattice (elastic anisotropy, nonlinear drag, and low crystallographic symmetry) and extrinsic physics that are due to defects other than dislocations (solute, vacancies, precipitates, and grain boundaries). For each of these classes of physics, we first discuss the conditions under which they are relevant, followed by an examination of the fundamental ways in which the behaviors of dislocations are affected by the physics, and finally a presentation of the methods that have been developed for incorporating the physics in DDD.

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Response of a Surface with Extensional and Bending Elasticity

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Abstract

Recent experiments and theory have shown that for a broad class of soft materials, their surface stress can strongly influence mechanical behavior. As an example relevant to wetting, a line force applied to the surface of an elastic substrate is locally, over a distance called the elastocapillary length l_c (ratio of surface stress and elastic modulus), supported mostly by surface stress. Surface stress also flattens surface profiles. In most cases studied the dominant actor is the surface stress and its possible variation with surface strain, i.e., its surface elasticity. However, there are other instances, such as lipid bilayer interfaces between aqueous phases, or thin oxidized layers on soft silicones, that also resist bending deformation. We will present experimental evidence for a system comprising a partially oxidized PDMS surface in which surface stress, in-plane elasticity, and surface bending, all matter. Specifically, we study the effect of such a surface on flattening of a patterned PDMS film. We will also discuss the response of a half-space to a line force when the surface carries both a stress and resistance to bending. We find that surface bending regularizes the singular elastic stress field due to a line load. The local stress field in the substrate near the line load can often be separated into three regions. Bending dominates in region 1; surface stress dominates in region 2; classical elasticity in region 3.

Elastic networks with optimal mechanical properties

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Various elastic systems can be modeled as networks of interconnected beams. Examples include polymer gels, protein networks, crystal atomic lattices, granular materials, wood, and bones. On a length scale much larger than the typical beam length (“macroscopic” scale), such a network can be viewed as a continuous and homogeneous medium characterized by spatially constant elastic moduli. The relationship between the mechanical properties of elastic networks on a macroscopic level, and the details of their microstructures is the key to optimization and design of lightweight, strong, and tough materials. The continuum modelling of such discrete structures has a long history, going as back as the pioneering work of A. Cauchy and S. Poisson. The stiffness of such a system clearly depends on its density ϕ , defined as the volume of beams per unit volume of material. But for a given value of ϕ , it is also dramatically affected by the specific spatial arrangement of the elastic phase within the material. On dimensional grounds, the volumetric density of strain energy ε associated with a stretch-dominated deformation varies linearly with ϕ , while it scales as ϕ^2 for the deformation of a three-dimensional network dominated by the beam bending mode [1]. Thus, for the low-density materials considered here ($\phi \ll 1$), a structure deforming primarily through the beam stretching mode is usually much stiffer. However, the constant of proportionality between ε and ϕ still varies significantly among stretched-dominated networks.

In this talk, I will present our numerical and theoretical investigations on the relation between the macroscopic mechanical response of a network under small strain condition and the details of its microstructure. I will stress that this relation is not straightforward. For instance, networks with identical node connectivity can have different stiffness. Conversely, networks with very different microstructures can present identical mechanical response. Moreover, contrary to a common belief, stiffness is not directly related to the triangulation of the mesh, and a fully triangulated network may have a lower stiffness than a network with few triangulated units. I will then show the existence of a class of isotropic networks which are stiffer than *any* other ones with the same symmetry, density and elastic phase [2, 3]. The elastic moduli of these optimal elastic networks can be calculated explicitly. They constitute upper-bounds which compete or improve the well-known Hashin-Shtrikman bounds. I will establish a general set of criteria (which turn out to be necessary and sufficient conditions) that allow to identify these networks.

Relation between the mechanical and transport properties of these networks will also be discussed. Finally, examples of such networks with periodic arrangement are presented, in both two and three dimensions.

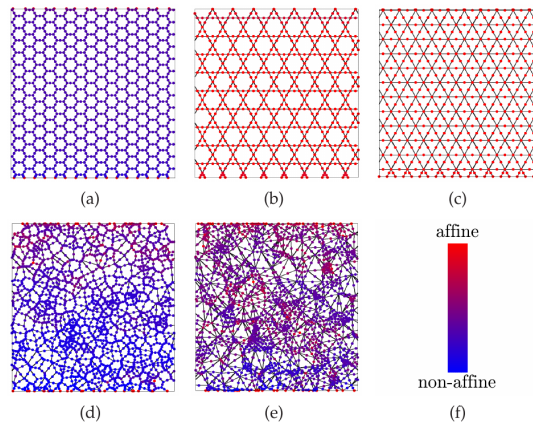


Figure 1. Affinity measure $m_i = \|\bar{\mathbf{u}}_i - \mathbf{u}_i\|/(\gamma L)$ of the displacement field for various 2d isotropic networks subject to horizontal shear strain γ : (a) hexagonal network; (b) kagome network; (c) triangular network; (d) Voronoi network; (e) Delaunay network; the (logarithmic) calibration bar (f) shows the coloration scheme used for the nodes i : they are coloured in red if their displacements match the affine displacement field, and blue in the opposite case.

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Geometric incompatibilities and bending of ultra-thin films

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Pre-stressed multi-layer materials minimize elastic energy through bending. While the phenomenon is well-understood for straight ribbons it is only partially studied in complex geometries. At the nanoscale, self-organisation through relaxation opens new fabrication methods for complex three-dimensional objects. We illustrate some peculiarities of the pre-stress in ultra-thin elastic bi-layers from the experimental viewpoint and discuss several open issues related to the relaxation of more complex geometries.

Atomic-Scale Modelling of Screw Dislocation Mobility in Zr and Ti

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Titanium and zirconium have a close plastic behaviour arising from their hexagonal close-packed crystallography and from their similar electronic structure. In particular, plasticity in these two transition metals is controlled by screw dislocations gliding in the prism planes, with cross-slip in the first-order pyramidal planes or in the basal planes activated at high enough temperature and a strong hardening associated with O addition. We use atomistic simulations relying on ab initio calculations and on empirical interatomic potentials to study core properties of the screw dislocations and their mobility in both metals. These calculations show that screw dislocations may adopt different cores that are dissociated either in a prism or in a pyramidal plane. The prismatic core easily glides in its habit plane, whereas the pyramidal core needs to overcome an important energy barrier to glide. The prismatic glissile core is the most stable in Zr, but the dislocation ground state in Ti corresponds to the pyramidal core. As a consequence, dislocation glide is easy and confined in the prismatic planes at low temperature in pure Zr, whereas a locking-unlocking mechanism operates in Ti where the locked periods correspond to a slow and limited glide in pyramidal planes and the unlocked periods to a rapid and extended glide in prismatic planes, in agreement with in situ TEM straining experiments. Finally, we study the interaction of interstitial solute atoms, in particular oxygen, with these different configurations of the screw dislocation. Ab initio calculations evidence a strong repulsion with the oxygen repelling the stacking fault ribbon, thus inducing dislocation cross-slip and leading to the production of jogs on screw dislocation. This short-range interaction mechanism is responsible for the strong hardening associated with oxygen addition.

A Three-dimensional Polycrystal Plasticity Model with Grain Boundary Evolution

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At high temperatures, grain boundaries are not static entities but can migrate in response to thermo-mechanical forces brought about by temperature, external stresses, and the internal microstructure. This gives rise to a wide array of phenomena where grain boundaries interact dynamically with bulk dislocation structures, such as during dynamic recovery, recrystallization, grain growth, etc.

In this work, we present a recently developed three-dimensional polycrystal plasticity model driven by a single energy functional that captures both grain boundary (GB) energy and bulk elastic energy under deformation. The model furnishes a framework for representing grain boundaries using geometrically necessary dislocations (GNDs), such that the GB energy is constructed in terms of these GND densities. Moreover, in contrast to previous phase-field approaches for grain boundary evolution, our model's energy functional is described on the entire five-dimensional grain boundary space, allowing the consideration of both misorientation and inclination in the grain boundary energy. We demonstrate the richness of the model by simulating coupled grain boundary motion, grain rotation and shrinkage, dynamic polygonization and anisotropic effects such as GB faceting.

On yield in amorphous solids under periodic shear

Turab Lookman (Los Alamos National Laboratory)

A fundamental problem in the physics of amorphous materials is understanding the transition from reversible to irreversible plastic behavior and its connection to yield. I will discuss how amorphous solids under periodic shear undergo a transition from repetitive, predictable behavior to chaotic, irregular behavior as a function of the strain amplitude. In both the periodic and chaotic regimes, localized particle rearrangements are observed. We associate the point of transition from repetitive to chaotic behavior as a dynamical transition as well as a critical point where avalanche sizes associated with displacements diverge.

Morphology and mechanical behaviour of fibrous materials as revealed via micro-computed tomography

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The last decade saw a rapid growth of μ CT applications for textiles and textile composites. This growth was in line with the fast development of the μ CT hardware. The X-ray computer tomography apparatus, from desktops to synchrotron-based instruments, allow the registration of 3D images with submicron and micron resolution, with a contrast sufficient to distinguish organic and carbon-based materials (like polymer matrices and carbon fibres) and the image post-processing is effective in eliminating the imaging artefacts. The micro-CT with a suite of software tools for quantification of the image is becoming a common tool in advanced experimental mechanics of fibrous and composite materials. Beautiful 3D images, however, are not used in full as the virtual representation of the internal structure of textiles and textile composites, because the voxel structure of the image lacks directionality, which is the paramount local characteristic of fibre reinforced composites and fibrous assemblies.

Description of a textile composite microstructure involves identification of individual yarns/fibrous plies in the textile reinforcement and definition of local parameters of fibrous geometry: local fibre directions, local fibre volume fraction and description of the amount and morphology of voids. The methods and the software, developed in KU Leuven (*VoxTex*) analyse X-ray micro-computed tomography (micro-CT) images, producing a description of the textile microstructure as an array of volume elements (voxels), each element carrying information of the fibre directions and fibre volume fraction in it. Apart from that, the content and morphology of void contents and the voids morphology in textile composites are characterised. The methods are based on a two-parameters analysis of the image: local grey scale value and anisotropy, defined via the structure tensor of the grey scale field.

Starting from the quantified micro-CT image, a number of questions related to the fibrous morphology and mechanics of the fibre reinforced material can be addressed. Morphological problems include fibre misalignment, regularities and irregularities of the internal structure in composites, created by different manufacturing processes, from random chopped strands layup to automated fibre placement. Transformation of a voxel analysis into a finite element model allows prediction of damage initiation and development in 2D and 3D woven carbon reinforced composites, and can be used for precise data reduction in an impregnated fibre bundle test, accounting for the structure of the bundle.

Heterogeneous mechanics of polymers in the glass transition regime, consequences and modeling.

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Since more than 20 years, the specific dynamics of glassy materials in their glass transition regime has been identified. The following picture has been established. A glass can be considered as a tiling of nanometric domains, each exhibiting its own dynamics that can differ from its neighbors by typically 4 orders of magnitude. We have begun a few years ago to investigate the consequences of this so-called “heterogeneous dynamics” on the mechanical properties of polymer glasses near their glass transition. For that we have developed a stochastic Finite elements model that is based on the physics of polymer glasses and that is able to describe quantitatively the linear viscoelasticity of polymer glasses. We will show that this model allows not only predicting quantitatively the linear visco-elasticity, but also the shift of dynamics observed in confined polymer and the onset of non-linearities in the glass transition regime.

Role of dynamical heterogeneities on the viscoelastic spectrum of polymers: a stochastic continuum mechanics model

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Role of Dynamical Heterogeneities on the Mechanical Response of Confined Polymers, R. J. Masurel, P.

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Structure and mechanics of self-organized fibrous materials with inter-fiber adhesion

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Abstract

Fibrous networks with random structure are ubiquitous in our everyday lives. From biological structures, such as the cellular cytoskeleton and connective tissue, to man-made structures including rubber, paper and non-wovens, fibrous materials are the structural building block of many material systems. The reasons for this prevalence are: (a) networks are lightweight structures, (b) are non-continuum materials, (c) and exhibit large sensitivity to structural controllable parameters. Adhesion plays an important role in the mechanics of nanoscale fibres. This is due to the fact that adhesion, as well as other surface forces, such as capillary forces, organize the fibres in bundles and contribute to the energy balance of the ensemble of fibres.

In this work we study the effect of inter-fibre adhesion on the self-organization of the ensemble of filaments as well as the mechanical behaviour of the network. We study networks in which fibres are cross-linked and networks in which the fibres are not cross-linked. In the first case, the connectivity of the network is preserved as the network organizes under the effect of adhesion. In the second case fibres form bundles which then re-organize in a network.

We observe that non-cross-linked networks may take one of three possible configurations: they may remain locked in the as-deposited state, may evolve under the action of inter-fibre adhesion to form “cellular networks”, or may disintegrate as a result of excessive bundling. We produce a phase diagram which indicates what structure is favoured by a certain selection of network parameters (density, fibre bending and axial stiffness, strength of inter-fibre adhesion) [1,2].

Cross-linked networks also self-organize, but to a much smaller extent due to the topological constraints imposed by the cross-links. This leads to significant volume reduction. The mechanical behaviour of these networks is very different from that of the same network without adhesion. They exhibit both softening and stiffening under uniaxial stretch or shear. Once again, we establish the relation between network parameters and various aspects of the mechanical behaviour [3].

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Rare events and uncertainty quantification in crystal plasticity

Stefanos Papanikolaou

Rare, abrupt events are common in plastically deforming crystals at small scales. Despite the spatial self-averaging of large samples, these abrupt events may control the behavior of sub-micron representative volumes in multiscale modeling or in delicate mechanical probes such as nanoindentation. The very origin of abrupt events is the presence of quenched disorder either in the form of inclusions or parent dislocation configurations. In this talk, I will discuss the form of such disorder distributions, their role in the compressive yield mechanical properties and also, possible ways of measuring them.

Liquid forces on elastic structures: capillarity prevails at small scales

Sebastien Neukirch

At small scales, liquid droplets develop capillary forces strong enough to deform thin elastic structures. For example, upon compression of its ends, a spider capture silk thread spontaneously bends and spools inside the water droplets sitting on it. This elasto-capillary coiling provides the hybrid fiber (thread+droplet) with a liquid-solid mechanical response. Indeed, capillary forces compressing the thread inside the droplet confer a constant tension to the system, even in a globally compressed configuration. This behavior is characteristic of liquid soapy films.



This one-dimensional liquid-solid fiber is now extended to a two-dimensional version: the liquid infused nano-fibrous membrane. Here, wrinkling of the membrane inside the liquid film provides a novel tightening self-storing mechanism; the hybrid membrane adapts its surface area even when compressed, always remaining under tension. The cohabitation of liquid interfaces and the folding of intrinsically inextensible fibers gives rise to interesting behaviors and allows us to re-visit classic objects such as liquid (and solid!) bubbles and catenoids.

Shear band development in a planar single crystal under dynamic loading conditions

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Shear band development in single crystals under quasi-static loading conditions has been investigated in a number of studies. On the other hand, there have been relatively few analyses of shear band development in single crystals under dynamic loading conditions. To explore dynamic effects, plane strain compression of a rectangular block is used as a model problem to investigate shear band development in a planar single crystal. The crystal geometry is idealized in terms of a planar double slip model and the material is characterized by a simple nonlocal crystal plasticity theory that does not require additional boundary conditions. The initial/boundary value problem is formulated accounting for finite strains and rotations, and material rate sensitivity. An initial material inhomogeneity is used to trigger shear band initiation. The focus is on the interaction of material rate sensitivity and the material length scale in affecting the shear band width and the shear band evolution.

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Title: Compression of fibrin and CNT networks viewed as a phase transition

Abstract: Fibrin is a polymer made from the blood plasma precursor protein fibrinogen. Fibrin forms a porous network of branching fibers that provides the scaffold of protective hemostatic blood clots and pathological obstructive thrombi. The structure and properties of the fibrin, including mechanical resilience to forces of blood flow and clot contraction, determine the course of pathological conditions, such as coronary heart disease, ischemic stroke, or bleeding. Hence, the tensile and shear behaviors of fibrin networks have been studied for a long time. However, compression of these networks remains largely unexplored. By characterization of the network structure with synchronous measurement of the fibrin storage and loss moduli at increasing degrees of compression, we show that the compressive behavior of fibrin networks is similar to that of cellular solids. A non-linear stress-strain response of fibrin consists of three regimes: 1) an initial linear regime, in which most fibers are straight, 2) a plateau regime, in which more and more fibers buckle and collapse, and 3) a markedly non-linear regime, in which network densification occurs by bending of buckled fibers and inter-fiber contacts. As such fibrin networks under compression have similar stress-strain curves to networks of carbon nanotubes (CNT) as well as macroscopic foams. Importantly, the spatially non-uniform network deformation included formation of a moving "compression front" along the axis of strain, which segregated the fibrin into compartments with different fiber densities and structure. This too has been seen in networks of carbon nanotubes as well as metallic foams. We model the mechanics of both types of network using a continuum theory of phase transitions and analytically predict (a) the storage and loss moduli for fibrin, and (b) the response of CNT networks to nanoindentation which are in good agreement with the experimental data. Our models could therefore aid the design of fibrin based biomaterials and CNT forests to have engineered mechanical properties, and guide further understanding of their behavior under large deformations.

Surface effects on elastic structures

Benoît Roman (ESPCI, Paris)

Although surface forces are not in general relevant to large scale objects, they may become dominant at small scale and deform or destroy micro-structures. I will show how surface effects can have important mechanical consequences on several examples: The surface tension of a liquid may deform an elastic object, as evidenced using experiments at a macroscopic scale. I will also describe how electrostatic surface effects may drive a controllable mechanical deformation.

Dislocation vibrations reveal the interplay between long-range elasticity and short-range line tension

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Dislocation vibrations are central to set the timescale of thermally activated processes such as cross-slip or precipitate by-pass. Historically, Friedel and then Granato and Lüke modeled dislocation vibrations using a simple line tension model. However, this approach neglects the long-range elastic interactions along the dislocation line. The latter can be modeled using the non-singular theory of dislocation as done in Dislocation Dynamics models, but this approach requires to define a smearing parameter. To obtain this parameter quantitatively, we used Molecular Dynamics simulations to determine the finite temperature vibrational spectrum of conventional $\frac{1}{2}\langle 110 \rangle \{111\}$ dislocations of different character in face-centered cubic Aluminum. The spectra are related through the equipartition theorem to the energetics of a dislocation subjected to a sinusoidal perturbation, which we express analytically using the non-singular theory of dislocations for the long-range interactions and a line tension model to represent the short-range core contribution. We show that the vibrational spectra cannot be fitted with elasticity or line tension alone, but we obtain an accurate fit combining both, with a smearing parameter and a core line tension, which depend on the dislocation character and the temperature. We discuss these values in comparison with static calculations on straight dislocations and we also explore to what extent these parameters are transferable and can be used, for example to predict quantitatively the bow-out of a dislocation between obstacles.